$\Rightarrow \qquad \frac{2}{9}\left[\frac{9}{2}\right] = 1$

It is to be noted that (i) $f(x) \geq 0, \forall x$ (ii) $\int_{-\infty}^{\infty} f(x)dx = 1$

Hence, $f(x)$ is a p.d.f.

9.4 Distribution function and its properties

We get the probability of a given event at a particular point. If we want to have the probability upto the point we get the probability $P(X \leq x)$. This type of probability is known as probability mass function. We can also find how the probability is distributed within certain limits. [$P(X < x)$ or $P(X > x)$ or $P(a < x < b)$].

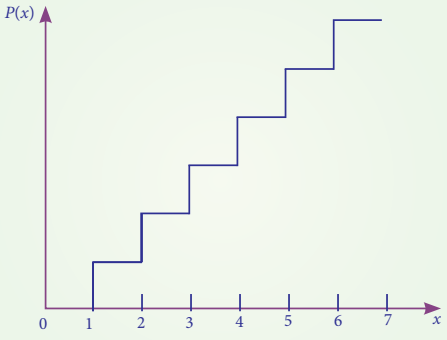


Fig.9.5

9.4.1 Distribution Function for discrete random variable

Definition: Let X be a random variable ,the cumulative distribution function (c.d.f) of a random variable X is defined as $F(x) = P(X \leq x), \forall x$. It is called simply as distribution function.

9.4.2 Properties:

(i) $0 \leq F(x) \leq 1, \ \forall x$, (non –negative)

(ii) $F(x) \leq F(y), \ \forall x < y$, (non –decreasing)

(iii) $\lim_{h \to 0} F(x+h) = F(x), \ \forall x$, ($F(x)$ is right continuous)

(iv) $F(-\infty) = \lim_{x \to -\infty} F(x) = 0$

(v) $F(\infty) = \lim_{x \to \infty} F(x) = 1$

NOTE

In the case of discrete random variable X

(i) $F(x) = \sum_{r=-\infty}^{x} P(x = r)$

(ii) $P(X = x) = F(x) - F(x-1)$

9.4.3 Distribution Function for continuous random variable

(i) $F(x) = \int_{-\infty}^{x} f(x)dx$

(ii) $f(x) = F'(x)$

(iii) $P(a < X < b) = P(a \leq X < b)$

$= P(a < X \leq b) = P(a \leq X \leq b)$

$= \int_a^b f(x)dx$

That is pdf can be obtained from distribution function by differentiating the distribution function and the distribution function can be obtained from pdf by integrating the pdf over the given range